Two-slack window-covering words

A triangular last-occurrence automaton for OEIS A242322

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Abstract

For integers $n, k \geq 1$, OEIS A242322 counts canonical words of length $n+k+2$ over $\{0, 1, \ldots, k\}$ in which every consecutive block of length $k+3$ contains every symbol. We encode a valid prefix by the two missing values among the possible ages of the most recent symbol occurrences. This gives an exact transfer matrix A_k with

$$\binom{k+2}{2}$$

states, explaining the triangular column orders observed in the OEIS entry. If w_k is an explicit initial vector, then

$$T(n,k) = w_k^{\mathsf{T}} A_k^{n-1} \mathbf{1}, \qquad \sum_{n \geq 1} T(n,k) x^n = x w_k^{\mathsf{T}} (I - x A_k)^{-1} \mathbf{1}.$$

We prove that the least homogeneous constant-coefficient scalar recurrence order is exactly $\binom{k+2}{2}$ for every k, even for recurrences required to hold only eventually. The proof combines a boundary generating-polynomial identity with a unimodular time-reversal form. We also prove the stable-row formula

$$T(n,k) = \frac{a_n}{8}k^4 + \frac{b_n}{12}k^3 + \frac{c_n}{8}k^2 + \frac{12n+19}{12}k + 1 \qquad (k \geq \max\{1, n-4\}),$$

where

$$\begin{aligned} a_n &= 4 \cdot 3^{n-1} - 2^{n+1} + 1, \\ b_n &= -4 \cdot 3^n + (6n+9)2^n - 6n - 1, \\ c_n &= 4 \cdot 3^{n-1} + (12-4n)2^n + 4n^2 - 5. \end{aligned}$$

Thus the previously empirical triangular recurrences and quartic stabilization have a common finite-state explanation.

1 Introduction

Let $\Sigma_k = \{0, 1, \ldots, k\}$. A word over Σ_k is *canonical* if its new symbols are introduced in the order $0, 1, \ldots, k$; equivalently, it is a restricted-growth word. Define

$$T(n,k) = \#\left\{ w \in \Sigma_k^{n+k+2} : \begin{array}{l} w \text{ is canonical, and} \\ \text{every factor of length } k+3 \text{ contains } \Sigma_k \end{array} \right\}. \tag{1}$$

This is the table A242322 [1]. Its entry records empirical column recurrence orders

$$3, 6, 10, 15, 21, 28$$

for $k = 1, \ldots, 6$, and empirical quartic formulas for its first fifteen rows. The orders are the triangular numbers $\binom{k+2}{2}$.

The main point of this paper is that the triangular numbers are not a numerical coincidence. At any time, record the age of the most recent occurrence of each symbol. A valid prefix has $k+1$ distinct ages among the $k+3$ possible ages $0, 1, \ldots, k+2$, and age zero is always present. Exactly two positive ages are therefore missing. Those two missing ages form a complete state description.

This produces a transfer system substantially smaller than the literal suffix automaton. It also retains enough structure to evaluate all fixed rows in the stable range. The resulting proofs use only finite-state enumeration, exact linear algebra, and finite sums.

The revised version also proves that the triangular order is minimal for every alphabet size. This is stronger than the characteristic recurrence bound: an exact transfer matrix can have states that cancel from its scalar generating function. Theorem 4.4 excludes every such cancellation by proving both cyclicity of the terminal vector and observability by the initial vector.

2 Relabeling and last-occurrence ages

Put

$$q = k + 1, \qquad r = k + 2 = q + 1.$$

The alphabet has $q = r - 1$ symbols and the required window length is $r + 1$.

Lemma 2.1 (Relabeling orbits)**.** *Every valid labelled word uses all q symbols. Relabeling by the symmetric group S_q acts freely, and every orbit contains exactly one canonical word. Consequently, canonical words may be counted as labelled words divided by $q!$.*

Proof. The first window contains every symbol. A relabeling that fixes the word therefore fixes all symbols, so the action is free. Relabeling the symbols by their order of first appearance produces the unique canonical representative of each orbit. □

For a valid prefix ending at position t, let

$$\alpha_t(s) = t - \max\{u \le t : w_u = s\}$$

be the *age* of symbol s. The ages are distinct, age zero is present, and validity is equivalent to $\alpha_t(s) \le r$ for every symbol.

Lemma 2.2 (Triangular state space)**.** *After the first complete window, the age set has the form*

$$\{0, 1, \ldots, r\} \setminus \{i, j\}, \qquad 1 \le i < j \le r.$$

Thus the quotient state space is

$$\mathcal{E}_r = \{(i, j) : 1 \le i < j \le r\}, \qquad |\mathcal{E}_r| = \binom{r}{2} = \binom{k+2}{2}.$$

Proof. There are $q = r - 1$ symbols, hence $r - 1$ distinct ages. They lie among the $r + 1$ integers $0, \ldots, r$, and zero occurs as the age of the final symbol. Their complement therefore consists of exactly two positive integers. □

Lemma 2.3 (Transitions)**.** *The successors of $(i, j) \in \mathcal{E}_r$ are as follows.*

1. *If $j < r$, the unique successor is $(i + 1, j + 1)$.*

2. If $j = r$, put $v = i + 1$. The successors are all pairs incident with v:

$$\{(\min\{v, u\}, \max\{v, u\}) : 1 \le u \le r,\ u \ne v\}.$$

Every listed edge is realized by exactly one appended symbol.

Proof. If $j < r$, age r is present. Its symbol must be appended, since otherwise its age would become $r + 1$. Resetting that age to zero and increasing all other ages shifts the two missing ages to $i + 1, j + 1$.

If $j = r$, all current ages are at most $r - 1$, so any symbol may be appended. Its current age a ranges over $\{0, \dots, r - 1\} \setminus \{i\}$. After the update, the missing ages are $i + 1$ and $a + 1$. As a varies, $a + 1$ takes every value in $\{1, \dots, r\} \setminus \{i + 1\}$ exactly once. □

Let A_k be the $\binom{r}{2}$ square matrix indexed by $\mathcal{E}_r$, with

$$(A_k)_{e,f} = \begin{cases} 1, & e \longrightarrow f \text{ in lemma 2.3,} \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 2.4 (Initial vector)**.** *The number of canonical first windows whose terminal age state is (i, j) is*

$$w_k(i, j) = i(j - 1).$$

Proof. A first window has length $r + 1$ and contains all $q = r - 1$ symbols. Fix the set of last-occurrence ages, namely $\{0, \dots, r\} \setminus \{i, j\}$. Assigning the q labels to these last-occurrence positions contributes $q!$. At the non-last position of age i, its symbol may have any of the i last occurrences of smaller age. At age j, there are $j - 1$ choices, since age i is not a last-occurrence position. These choices are independent. There are therefore $q!i(j - 1)$ labelled windows. Divide by $q!$ using lemma 2.1. □

3 Exact transfer formula and column recurrences

Theorem 3.1 (Exact transfer enumeration)**.** *For all $n, k \ge 1$,*

$$T(n, k) = w_k^{\mathsf{T}} A_k^{n-1} \mathbf{1}, \tag{2}$$

where $w_k(i, j) = i(j - 1)$ and $\mathbf{1}$ is the all-one column vector. Hence

$$\mathcal{T}_k(x) := \sum_{n \ge 1} T(n, k) x^n = x w_k^{\mathsf{T}} (I - x A_k)^{-1} \mathbf{1}. \tag{3}$$

Proof. Lemma 2.4 counts the possible first windows by state. Lemma 2.3 gives a one-to-one correspondence between appended symbols and state edges. A word counted by $T(n, k)$ has $n - 1$ transitions after its first window, and its final state is unrestricted. This proves (2); summing the matrix geometric series proves (3). □

Define

$$D_k(x) = \det(I - x A_k) = \sum_{\ell=0}^{N} d_{k,\ell} x^\ell, \qquad N = \binom{k + 2}{2}.$$

Proposition 3.2 (Unimodularity)**.** *The matrix A_k is unimodular. In particular, $d_{k,0} = 1$, $|d_{k,N}| = 1$, and D_k has degree exactly N.*

Proof. Order first the rows (i, j) with $j < r$, then the boundary rows (i, r). Order first the columns (u, v) with $2 \le u < v \le r$, then the columns $(1, v)$.

The interior transition $(i, j) \mapsto (i + 1, j + 1)$ is a bijection from the interior rows to the first group of columns. The corresponding block is a permutation matrix, and the upper-right block is zero. In boundary row (i, r), the only target in the second column group is $(1, i + 1)$. The lower-right block is therefore an identity matrix. The reordered matrix is block lower triangular with unimodular diagonal blocks. □

Corollary 3.3 (Triangular-order recurrence)**.** *For every $k \ge 1$, the column $T(n, k)$ satisfies*

$$\sum_{\ell=0}^{N} d_{k,\ell} T(n - \ell, k) = 0 \qquad (n \ge N + 1), \qquad N = \binom{k+2}{2}. \tag{4}$$

Thus A242322 has a characteristic recurrence of the triangular order displayed empirically on OEIS.

Proof. Cayley–Hamilton gives $\sum_{\ell=0}^{N} d_{k,\ell} A_k^{N-\ell} = 0$. Multiplication by the initial and terminal vectors gives (4). The final coefficient is nonzero by proposition 3.2. □

Remark 3.4. Unimodularity alone does not prove scalar minimality. The stronger statement that N is the least possible order for every k is proved separately in theorem 4.4; it is not inferred from finite computations.

For example, exact characteristic-polynomial calculation gives

$$\begin{aligned} D_1(x) &= 1 - x - x^2 - x^3, \\ D_2(x) &= 1 - x - 2x^2 - 2x^3 - 2x^4 + x^5 + x^6, \\ D_3(x) &= 1 - x - 2x^2 - 3x^3 - 5x^4 - 6x^5 + x^6 + x^7 + x^9 + x^{10}. \end{aligned}$$

The first two agree with the recurrences printed on A242322.

4 Minimality of the scalar recurrence

Throughout this section, write $A = A_k$, $w = w_k$, $u = \mathbf{1}$, and $m = r - 1 = k + 1$. We work over $\mathbb{C}$; excluding a shorter recurrence over $\mathbb{C}$ also excludes one over $\mathbb{Q}$ or $\mathbb{Z}$. The proof does not use a conjectured closed formula for a determinant.

4.1 Boundary reduction

Let $b(i, j) = [j = r]$, where brackets denote an indicator. The row sums in lemma 2.3 give

$$(A - I)u = kb. \tag{5}$$

Use gap and boundary-distance coordinates

$$d = j - i, \qquad s = r - j, \qquad 1 \le d \le m, \quad 0 \le s \le m - d.$$

The boundary coordinates are $(d, 0)$. At an interior coordinate, the equation $(I - zA)f = g$ reads

$$f_{d,s} - z f_{d,s-1} = g_{d,s} \qquad (s \ge 1).$$

Consequently its interior block is triangular with diagonal ones and can be eliminated for every z. With zero interior forcing, this elimination gives $f_{d,s} = z^s f_{d,0}$. The boundary Schur complement is

$$M_m(z) = I - K_m(z), \qquad K_m(z)_{d,e} = [e \le m+1-d]z^d + [e < d]z^{d-e}. \tag{6}$$

To verify (6), start at $(r-d, r)$ and put $v = r-d+1$. A successor is the pair incident with v and a chosen $a \neq v$. If $a < v$, its gap is $e = v - a$, its boundary distance is $d-1$, and its next return to the boundary takes d steps; the possible gaps are $1 \le e \le m+1-d$. If $a > v$, its gap is $e = a - v$, its boundary distance is $d-1-e$, and the return takes $d-e$ steps; here $1 \le e < d$. These are exactly the two terms in (6), including the multiplicity when both types have the same gap.

Lemma 4.1 (Boundary nullvectors)**.** *Suppose $m \ge 3$ and $z \neq 0$. If a row vector $v \in \mathbb{C}^m$ satisfies $vM_m(z) = 0$ and $v\mathbf{1} = 0$, then $v = 0$.*

Proof. Write $v = (v_1, \dots, v_m)$ and $V(t) = \sum_{d=1}^m v_d t^d$. The eth component of $vM_m(z) = 0$ is

$$v_e = \sum_{d \le m+1-e} v_d z^d + \sum_{d>e} v_d z^{d-e}. \tag{7}$$

Multiply by t^e and sum over e. The finite geometric sums give, in the rational-function field $\mathbb{C}(t)$,

$$V(t) = \frac{t}{1-t}\big(V(z) - t^{m+1}V(z/t)\big) + \frac{tV(z) - zV(t)}{z-t}.$$

After multiplying by $(1-t)(z-t)$ and collecting terms, this is

$$a(t)V(t) = c(t)V(z) + h(t)V(z/t), \tag{8}$$

where

$$a(t) = (1-t)(2z-t), \qquad h(t) = -t^{m+2}(z-t), \qquad c(t) = t(z+1-2t).$$

All divisions above are in the indeterminate t, so no assumption $t \neq 1$ or $t \neq z$ is imposed on a later evaluation of a polynomial identity.

Replace t by z/t in (8) and eliminate $V(z/t)$:

$$\big(a(t)a(z/t) - h(t)h(z/t)\big)V(t) = \big(c(t)a(z/t) + h(t)c(z/t)\big)V(z). \tag{9}$$

Direct multiplication gives

$$\begin{aligned} a(t)a(z/t) - h(t)h(z/t) &= \frac{z(1-t)(t-z)}{t^2}Q_z(t), \\ Q_z(t) &= -2t^2 + (4z+1-z^{m+2})t - 2z. \end{aligned} \tag{10}$$

Since $z \neq 0$ and the leading coefficient of Q_z is -2, the rational function in (10) is nonzero. Thus (9) implies

$$v \neq 0 \quad \Longrightarrow \quad V(z) \neq 0 : \tag{11}$$

if $V(z) = 0$, that equation forces $V(t) = 0$ in the integral domain $\mathbb{C}(t)$, so every v_d is zero.

The other factor in (9) is

$$\begin{aligned} c(t)a(z/t) + h(t)c(z/t) &= \frac{z(t-z)}{t}R_z(t), \\ R_z(t) &= (z+1-2t)(2t-1) + t^{m+1}\big((z+1)t - 2z\big). \end{aligned} \tag{12}$$

Cancel the nonzero rational function $z(t-z)/t^2$ to obtain the polynomial identity

$$(1-t)Q_z(t)V(t) = tR_z(t)V(z). \tag{13}$$

From (12),

$$R_z(1) = 0, \qquad R_z'(1) = (m-2)(1-z).$$

Differentiate (13) and evaluate at $t = 1$:

$$(2z - 1 - z^{m+2})V(1) = (m-2)(z-1)V(z). \tag{14}$$

There is no division by $Q_z(1)$, so the identity also applies when that number vanishes. If $v \neq 0$ and $v\mathbf{1} = V(1) = 0$, then (11), (14), and $m \geq 3$ force $z = 1$.

It remains to exclude a nonzero left nullvector of $M_m(1)$. Every row of $K_m(1)$ has sum m, so $M_m(1)\mathbf{1} = (1-m)\mathbf{1}$. A left nullvector therefore has $\sum_d v_d = 0$, since $m \geq 2$. Put $S_e = \sum_{d=1}^{e} v_d$, with $S_0 = S_m = 0$. Equation (7) at $z = 1$ becomes

$$2S_e - S_{e-1} = S_{m+1-e} \qquad (1 \leq e \leq m).$$

If $S_{j-1} = S_{m+1-j} = 0$, the equation with $e = j$ gives $S_j = 0$; the equation with $e = m+1-j$ then gives $S_{m-j} = 0$. Starting at $j = 1$ and proceeding inward proves that all S_e vanish. Hence every $v_d = S_d - S_{d-1}$ is zero, as required. □

Proposition 4.2 (Cyclicity)**.** *For every $k \geq 1$, the matrix*

$$C_k = [u, A_k u, \ldots, A_k^{N-1}u], \qquad N = \binom{k+2}{2},$$

is invertible.

Proof. First suppose $k \geq 2$. If the Krylov span of u were a proper invariant subspace, its annihilator would be a nonzero invariant subspace of the dual space. Over $\mathbb{C}$, it contains a nonzero left eigenvector ℓ:

$$\ell A = \lambda \ell, \qquad \ell u = 0.$$

Unimodularity in proposition 3.2 excludes $\lambda = 0$. Equation (5) gives $\ell b = 0$ because $k \neq 0$. Put $z = 1/\lambda$, so $\ell(I - zA) = 0$.

Order the boundary coordinates first and write

$$I - zA = \begin{pmatrix} B & C \\ D & E \end{pmatrix}, \qquad \ell = (v, h).$$

The interior block E is invertible, as shown above. Thus $h = -vCE^{-1}$, and the other block equation gives $v(B - CE^{-1}D) = vM_m(z) = 0$. Moreover, $v \neq 0$, since $v = 0$ would also give $h = 0$. The vector b has all-one boundary part and zero interior part, so $\ell b = 0$ says $v\mathbf{1} = 0$. This contradicts lemma 4.1, because $m = k+1 \geq 3$ and $z \neq 0$. The Krylov span is therefore the whole space; Cayley–Hamilton shows that the first N vectors already span it.

For $k = 1$, in the state order $(1,2), (1,3), (2,3)$,

$$A_1 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad C_1 = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 2 & 4 \end{pmatrix},$$

and $\det C_1 = 1$. □

4.2 Reversal form and scalar minimality

Lemma 4.3 (Unimodular reversal form)**.** *Define a matrix indexed by $\mathcal{E}_r$ by*

$$J_{(i,j),(a,b)} = [i+a \geq r+1][j+b \geq r+1] + [i+b \geq r+1][j+a \geq r+1]. \tag{15}$$

Then $J = J^{\mathsf{T}}$, $\det J = \pm 1$, $Ju = w$, and $A^{\mathsf{T}}J = JA$.

Proof. Symmetry is immediate. Reverse each column pair by writing $(a,b) = (r+1-q, r+1-p)$, where $p < q$. Its entry becomes

$$[i \geq q] + [i \geq p][j \geq q].$$

Both summands vanish unless (i,j) is componentwise at least (p,q), and the diagonal entry is one. In a linear extension of componentwise order, the column-reversed matrix is unit lower triangular. Hence $\det J = \pm 1$.

For fixed (i,j), summing (15) over unordered pairs is equivalent to counting ordered distinct pairs (a,b) satisfying $a \geq r+1-i$ and $b \geq r+1-j$. There are $ij - i = i(j-1)$ such pairs, which proves $Ju = w$.

Let $e_{i,j}$ denote a coordinate vector. The transition rule gives

$$Ae_{1,j} = e_{j-1,r}, \qquad Ae_{i,j} = e_{i-1,j-1} + e_{i-1,r} + e_{j-1,r} \quad (i \geq 2).$$

Put $L = r+2$. Substitution in (15) gives

$$\begin{aligned}(JA)_{(i,j),(a,b)} = {} & [i+a \geq L][j+b \geq L] + [i+b \geq L][j+a \geq L] \\ & + [i+a \geq L] + [j+a \geq L] + [i+b \geq L] + [j+b \geq L].\end{aligned}$$

This formula includes $a = 1$, because every term involving $i+a$ or $j+a$ then vanishes. The expression is symmetric under exchange of the two pairs. Thus JA is symmetric, which is $A^{\mathsf{T}}J = JA$. □

Theorem 4.4 (Minimal triangular order)**.** *For every $k \geq 1$, the least order of a homogeneous constant-coefficient scalar recurrence for $T(n,k)$ is exactly $N = \binom{k+2}{2}$. This remains true if the recurrence is required to hold only for all sufficiently large n.*

Proof. Let C_k be as in proposition 4.2, and define

$$O_k = \begin{pmatrix} w^{\mathsf{T}} \\ w^{\mathsf{T}}A \\ \vdots \\ w^{\mathsf{T}}A^{N-1} \end{pmatrix}.$$

By lemma 4.3, $O_k^{\mathsf{T}} = JC_k$. Both J and C_k are invertible, so O_k is invertible as well. For every integer $s \geq 0$, the shifted Hankel matrix

$$H_s = \big(T(s+a+b+1,k)\big)_{0 \leq a,b < N} = O_k A^s C_k$$

is invertible, using proposition 3.2.

If a scalar recurrence of order $d < N$ held eventually, choose s large enough that it applies to every entry in column d of H_s. That column would then be a fixed linear combination of the preceding d columns, contradicting invertibility. No smaller eventual order is possible. Corollary 3.3 supplies a recurrence of order N, proving the claim. □

5 Stable rows

We now evaluate (2) when the alphabet is sufficiently large relative to the number of transitions.

Use distance coordinates

$$a = r - j, \qquad b = r - i, \qquad 0 \le a < b \le r - 1.$$

Let $H_m(a, b)$ be the number of length-m walks starting at this state, with unrestricted endpoint, and put $G_m(b) = H_m(0, b)$. Then $H_0(a, b) = 1$, and the transition rules give

$$H_m(a, b) = H_{m-1}(a - 1, b - 1) \qquad (a > 0), \tag{16}$$

$$G_m(b) = \sum_{\substack{0 \le c \le r-1 \\ c \ne b-1}} H_{m-1}(\min\{b - 1, c\}, \max\{b - 1, c\}). \tag{17}$$

Lemma 5.1 (Boundary walks)**.** *Suppose $s \ge 1$ and $r \ge s - 1$. For every $1 \le b \le r - 1$,*

$$G_s(b) = \begin{cases} 2^{s-1}(r - 2) + 1, & b \ge s, \\ 2^{b-1}3^{p-1}r^2 + (2^{p-1} + 2^{s-1} - 5 \cdot 2^{b-1}3^{p-1})r \\ \qquad + 2^b 3^p - 2^s - 2^p + 1, & b < s, \quad p = s - b. \end{cases}$$

Proof. We use strong induction on s. Repeated application of (16) gives

$$H_m(a, b) = \begin{cases} 1, & m \le a, \\ G_{m-a}(b - a), & m > a. \end{cases}$$

Substitution into (17) yields, for $b < s$ and $p = s - b$,

$$G_s(b) = \sum_{e=1}^{b-1} G_{p+e}(e) + \sum_{e=1}^{r-b} G_p(e). \tag{18}$$

For $b \ge s$, the same decomposition has only the constant branch and gives

$$G_s(b) = r - 1 + \sum_{t=1}^{s-1} (2^{t-1}(r - 2) + 1 - 1) = 2^{s-1}(r - 2) + 1.$$

For the second sum in (18), the induction hypothesis and geometric sums give, for every $R \ge p - 1$,

$$\sum_{e=1}^{R} G_p(e) = R(2^{p-1}(r - 2) + 1) + U_p(r),$$

where

$$U_p(r) = (3^{p-1} - 2^{p-1})r^2 + (-5 \cdot 3^{p-1} + 3 \cdot 2^p - 1)r + 2 \cdot 3^p - 4 \cdot 2^p + 2.$$

The first sum in (18) uses the nonconstant induction branch. Summing its three coefficients over $e = 1, \ldots, b - 1$, then adding the preceding display with $R = r - b$, gives exactly the stated expression. The same calculation remains valid at the limiting value $r = s - 1$, where $R = p - 1$. □

Theorem 5.2 (Stable quartic rows)**.** *For $n, k \geq 1$ with $k \geq \max\{1, n-4\}$,*

$$T(n,k) = \frac{a_n}{8}k^4 + \frac{b_n}{12}k^3 + \frac{c_n}{8}k^2 + \frac{12n+19}{12}k + 1, \tag{19}$$

where

$$\begin{aligned} a_n &= 4 \cdot 3^{n-1} - 2^{n+1} + 1, \\ b_n &= -4 \cdot 3^n + (6n+9)2^n - 6n - 1, \\ c_n &= 4 \cdot 3^{n-1} + (12-4n)2^n + 4n^2 - 5. \end{aligned}$$

The uniform boundary cannot be lowered: at $(n,k) = (6,1)$ the true value is 149, whereas the polynomial gives 151.

Proof. Put $m = n - 1$. In distance coordinates, the initial weight from lemma 2.4 is

$$(r-b)(r-a-1).$$

Consequently,

$$T(n,k) = \sum_{0 \leq a < b \leq r-1} (r-b)(r-a-1)H_m(a,b). \tag{20}$$

The hypothesis $k \geq n-4$ is exactly $r \geq m-1$, so lemma 5.1 applies to every nontrivial term after (16) is iterated. Substitute that lemma into (20), split the sum at $b-a = m-a$, and use

$$\sum_{u=1}^{M} u = \frac{M(M+1)}{2}, \quad \sum_{u=1}^{M} u^2 = \frac{M(M+1)(2M+1)}{6}, \quad \sum_{u=1}^{M} 2^u = 2^{M+1} - 2.$$

Collecting powers of $k = r - 2$ gives, respectively,

$$\frac{a_n}{8}, \quad \frac{b_n}{12}, \quad \frac{c_n}{8}, \quad \frac{12n+19}{12}, \quad 1,$$

which proves (19). Direct enumeration at $(6,1)$ gives the final boundary statement. □

6 Verification, scope, and disclosure

No proof step depends on fitting numerical data. Two standard-library-only enumeration programs accompany the manuscript. The feasibility verifier implements an equality-pattern quotient, a raw labelled suffix DFA, and literal restricted-growth enumeration. The theorem verifier independently implements the age automaton, exact characteristic polynomials, the boundary-walk lemma, and the stable-row formula. Together they check:

1. the triangular state classification and every transition;
2. agreement of four enumeration routes;
3. the first 10×10 values displayed on A242322;
4. characteristic recurrences and unimodularity through $k = 10$;
5. the boundary-walk formula, including its limiting case;
6. the stable-row theorem by matrix walks through $n = 40$, and independently by direct summation of the boundary lemma through $n = 100$.

An additional standard-library-only symbolic checker verifies the reflection identities on Laurent polynomials, the boundary Schur complement, and the coordinate action of A_k for $m = 2, \ldots, 10$. These are regression checks on the algebra in the general minimality proof, not a finite-case substitute for theorem 4.4. Finite checks support the audit but do not replace the proofs.

The matrix formula is an exact generating function for every column. The paper proves that its denominator degree is the minimal scalar recurrence order for every k. It does not claim a closed coefficient formula for $D_k(x)$ or closed formulas for the controllability determinant or its Smith normal form; those stronger arithmetic formulas remain conjectural. The literature search found the empirical statements on A242322 but no prior proof of the age-state reduction; this is a qualified status report rather than an absolute priority claim.

AI systems were used during exploration, adversarial checking, and drafting. The mathematical evidence consists of the explicit arguments and deterministic verification programs, not AI output. This draft is marked for independent mathematical review.

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